

Naga Durga Jagadeesh Lanka

Mechanical Engineering | Design and Manufacturing | Robotics
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PROFESSIONAL SUMMARY

Mechanical engineer completing an MEng at the University of Cincinnati, with CAD, drawing, fit-check and prototype experience. Seeking manufacturing, mechanical design and simulation roles, with additional experience in technical AI review and agent building.

TECHNICAL SKILLS

Design: SolidWorks, AutoCAD, Creo, Onshape, GD&T, tolerance analysis, fabrication drawings, BOMs

Analysis: FEA, fit and interference checks, dynamics, DFM/DFA, prototype testing

Programming and Robotics: Python, MATLAB, ROS 2, LiDAR integration, Gazebo, SQL

Documentation: Technical reports, revision control, PLM, Excel, Word, PowerPoint, Power BI

WORK EXPERIENCE

Mechanical Designer | Remote Internship

May - Aug 2026

Zouka Solutions, Hyderabad, India

- Developed and revised CAD models and engineering drawings from team requirements, including dimensions, tolerances and basic GD&T.
- Prepared calculations, performed fit and interference checks, and incorporated feedback from a colleague and manager.
- Maintained BOMs, component lists and revision records; coordinated design updates with manufacturing and procurement contacts.

Graduate Research Assistant

Dec 2025 - Apr 2026

UPRISE Project, University of Cincinnati, Ohio

- Reviewed technical studies and maintained a literature tracker to compare findings, check data quality and support reports for an NSF-funded research team.

Design Engineer Intern

Jan - May 2024

GSR Engineering, Hyderabad, India

- Supported a user-driven floor scrubber combining dry vacuuming and wet mopping, including drivetrain sizing and fit checks.
- Prepared fabrication drawings and a BOM, and supported prototype tests with senior engineers.

Engineering Intern

May - Jun 2022

Oil and Natural Gas Corporation Limited, India

- Studied gas-compressor controls, lubrication systems, and trip and alarm settings.

EDUCATION

Master of Engineering in Mechanical Engineering

Expected Dec 2026

University of Cincinnati, Ohio

GPA: 3.2 | Coursework: Robotics, Micro and Nano Manufacturing, Quality Control, Robot Control and Design, Industrial Big Data and AI.

Bachelor of Technology in Mechanical Engineering

2024

National Institute of Technology Manipur, India

Thesis: User-Driven Floor Scrubbing Machine, covering chain and bevel gearing, free-body analysis, mechanical advantage and operating costs.

ENGINEERING PROJECTS

Servo Motor Mounting and Belt Tensioning Assembly

Sep 2026

Academic report extension of internship work

- Developed an AI-assisted report extension with simplified CAD, dimensioned drawings and structural checks, separate from company design work.
- Evaluated a 20 mm travel design and predicted plate deflection. Documented load/support assumptions and the need for physical alignment checks.

Robotic Work Cell for Turning Mechanism Assembly

Dec 2024

University of Cincinnati | Two-person design project

- Designed bearing, shaft and lid feeding approaches, selected sensors and prepared the robot-motion flowchart.
- Collaborated on assembly animation and presentation of the team's work-cell concept.

TurtleBot3 Reactive Obstacle Avoidance

Jan - Apr 2025

ROS 2, Python, LiDAR and Gazebo | Team project

- Contributed to LiDAR-based control using a front detection sector and a stop, wait and turn state machine.
- Tested the approach in simulation and on hardware; documented missed detections, false positives and unstable physical avoidance.

Advances in Nanotechnology in the Aerospace Industry

Sep - Dec 2024

University of Cincinnati | Team literature review

- Co-authored a review of carbon nanotubes, self-healing materials, riblets, nanosensors and lightweight structures for aerospace applications.

Vertical Axis Wind Turbine

2024

CAD, fabrication and field testing

- Developed CAD models and a BOM for a fabricated prototype, and collected field wind measurements to compare behaviour with design intent.

AI AND SOFTWARE EXPERIENCE

Reviewer

Jul 2025 - Present

Mercor

- Review technical AI coding evaluations against rubrics, annotations and test evidence, and provide specific revision feedback.

Agent Building and Evaluation

2026

Outlier | OpenClaw

- Built an agent in OpenClaw and worked on environment setup, task specifications, rubrics and trajectory review, including tool use and privacy boundaries.

Selected Technical Projects

2026

- Authored tasks for Handshake AI projects Insight, Neutron Mechanical Engineering, Artemis and Planck; developed technical reasoning and evidence-based evaluation material.
- Created mechanical reasoning evaluations for Lumiere and reviewed response quality, personalization and scoring consistency for Novel.
- Built playable games in Defold, Panda3D and Godot for Touchstone, with asset integration, task specifications, rubrics and QA.
- Built AUREN, an AI-assisted Windows voice-assistant prototype using Python, SQLite and a web UI, with local command routing and spoken replies.

CERTIFICATION

Certified Associate in Project Management (CAPM) | Project Management Institute